Timing a possible encounter

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate orbital period from an encounter time.

SUCCESS CRITERIA - I can...

1. Use a clearly stated period and time reference.
2. Check positions of both bodies at the same time.
3. Explain why geometry and uncertainty matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

At fixed mass and speed, circle radius becomes 3 times as large. Compare acceleration and net force using equations.

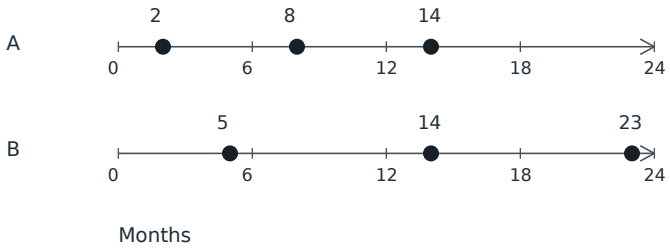
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $t_{arrival} = t_{start}$ + travel time.
An encounter needs matching position and time. A period T is one complete orbit. Use the supplied fictional timing model.

Worked reasoning: Two objects pass the same crossing point: A at months 2, 8, 14 and B at months 5, 14, 23. Only month 14 is a shared time in these lists.

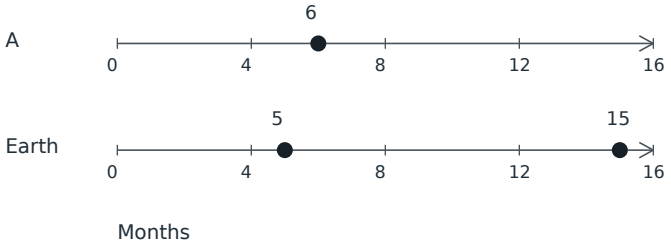
Your turn: Draw two timelines and connect the matching time. Explain why a shared location alone is insufficient.



Marzano check - circle: ① ② ③

03 Read and interpret the model

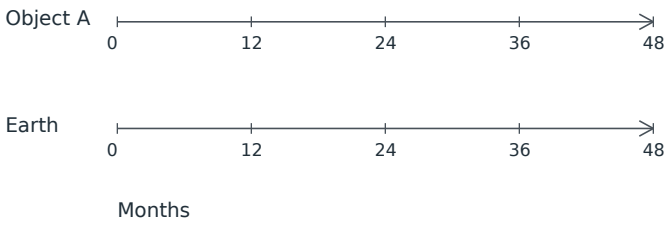
A simplified orbit model has two paths that cross at the same point. Object A arrives in 6 months. Earth arrives in 5 months and again in 15 months. Mark these times. Does sharing a crossing point establish an encounter? Explain.



Marzano check - circle: ① ② ③

04 Apply the physics

Toy crossings: asteroid at months 6, 24, 42; Earth at months 6, 18, 30, 42. Identify common crossing times and explain a limit of the model.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student turns a ±2-day timing range into “a 50% chance of impact.” Explain why that does not follow.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

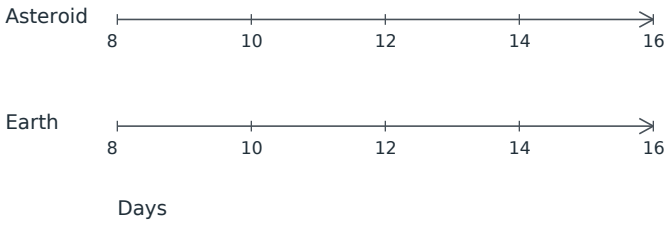
An object weighs 39.2 N where g=9.8 N/kg. Find its mass and its weight in a field of 6 N/kg.
Label the masses, field or force relevant to the question.



Marzano check - circle: 1 2 3

07 Exit check

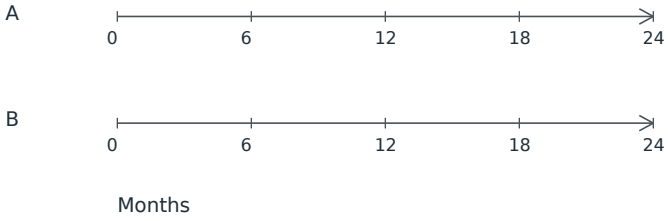
An asteroid’s crossing window is days 10-14; Earth’s is days 13-15. Find overlap and state what it does and does not establish.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A reaches a crossing at months 1+6 n; B at months 3+8 m, for nonnegative integers n, m. Find their first common crossing time and a limit of this timing model.



Marzano check - circle: 1 2 3